THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — MATHEMATICS · FULL SYLLABUS EXAMINATION

Session 2026-27 | CBSE / RBSE Pattern

Class: 9	Subject: Maths	Set 04	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper contains 38 questions and is divided into 5 sections — A, B, C, D and E.
2. In Section A, Q. 1–18 are multiple-choice questions and Q. 19–20 are Assertion–Reason based questions of 1 mark each.
3. In Section B, Q. 21–25 are very short answer (VSA) type questions carrying 2 marks each.
4. In Section C, Q. 26–31 are short answer (SA) type questions carrying 3 marks each.
5. In Section D, Q. 32–35 are long answer (LA) type questions carrying 5 marks each.
6. In Section E, Q. 36–38 are case-study based questions carrying 4 marks each with sub-parts of 1, 1 and 2 marks respectively.
7. All questions are compulsory. However, an internal choice is provided in 1 question of Section B, 2 questions of Section C and 2 questions of Section D. An internal choice is provided in all the 2-mark sub-parts of Section E.
8. Draw neat and clean figures wherever required. Take π = 22/7 wherever required, if not stated. Use of calculators is not allowed.
9. Write the serial number of each question before attempting it.

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. The abscissa of every point lying on the y-axis is: **1**

(a) equal to its ordinate (b) 0
(c) 1 (d) undefined

2. If (x − 2, y + 5) = (0, 0), then the value of x + y is: **1**

(a) −3 (b) 7 (c) −7 (d) 3

3. The midpoint of the segment joining (−6, 3) and (4, −9) is: **1**

(a) (−1, −3) (b) (−5, −6) (c) (1, −3) (d) (−1, 3)

4. The value of the polynomial 2x − 7 at x = −3 is: **1**

(a) 13 (b) −13 (c) −1 (d) −9

5. The zero of the polynomial 3x + 9 is: **1**

(a) 3 (b) −9 (c) 9 (d) −3

6. The side of an equilateral triangle is (2x − 1) cm. Its perimeter (in cm) is: **1**

(a) 8x − 4 (b) 6x − 3 (c) 2x − 3 (d) 6x − 1

7. Which of the following is a rational number? **1**

(a) √(49/16) (b) √7
(c) π (d) 0.12112111211112…

8. The decimal expansion of $\frac{7}{16}$ terminates after how many decimal places? **1**

(a) 5 (b) 3 (c) 4 (d) 2

9. The value of $|-\frac{2}{3}| \times |\frac{9}{4}|$ is: **1**

(a) $\frac{1}{6}$ (b) $\frac{8}{27}$ (c) $\frac{-3}{2}$ (d) $\frac{3}{2}$

10. On expanding $(2x + 5)^2$, we get: **1**

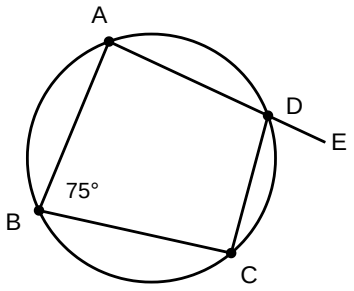
(a) $4x^2 + 25$ (b) $2x^2 + 20x + 25$ (c) $4x^2 + 20x + 25$ (d) $4x^2 + 10x + 25$

11. The factorisation of $1 - 49a^2$ is: **1**

(a) $(1 - 49a)(1 + a)$ (b) $(1 - 7a)^2$

(c) $(1 - 7a)(1 + 7a)$ (d) $(7a - 1)^2$

12. In the figure, ABCD is a cyclic quadrilateral and the side AD is produced to E. If $\angle ABC = 75°$, then the exterior angle $\angle CDE$ is: **1**



(a) 105° (b) 15° (c) 75° (d) 150°

13. The longest chord of a circle of radius 5 cm has length: **1**

(a) 2.5 cm (b) 20 cm (c) 5 cm (d) 10 cm

14. The area of a right-angled triangle with sides 6 cm, 8 cm and 10 cm, found using Heron's formula, is: **1**

(a) 24 cm^2 (b) 30 cm^2 (c) 48 cm^2 (d) 40 cm^2

15. The area of an isosceles triangle with sides 25 cm, 25 cm and 14 cm is: **1**

(a) 84 cm^2 (b) 175 cm^2 (c) 336 cm^2 (d) 168 cm^2

16. A number is picked at random from the numbers 1 to 25. The probability that it is a prime number is: **1**

(a) $\frac{9}{25}$ (b) $\frac{8}{25}$ (c) $\frac{10}{25}$ (d) $\frac{12}{25}$

17. A coin and a die are thrown together. The number of possible outcomes is: **1**

(a) 8 (b) 6 (c) 36 (d) 12

18. The sum of the first 10 natural numbers is: **1**

(a) 110 (b) 55 (c) 45 (d) 50

19. **Assertion (A):** The sequence 5, 8, 11, 14, … is an AP with common difference 3. **1**
Reason (R): In an AP the difference between consecutive terms is constant.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

20. **Assertion (A):** The area of a triangle with sides 7 cm, 8 cm and 20 cm can be found using Heron's formula. **1**
Reason (R): The sum of any two sides of a triangle is greater than its third side.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. A plumber charges ₹150 as visiting charge and ₹75 for every hour of work. Write the expression for the total charge for h hours and find it for 4 hours. **2**

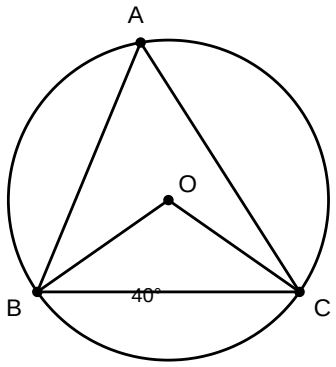
22. Without actual division, show that $\frac{51}{1600}$ has a terminating decimal expansion and state the number of decimal places. **2**

OR

Find two rational numbers between 0.1 and 0.2.

23. Using a suitable identity, evaluate 1002 × 998. **2**

24. In the figure, O is the centre of the circle and ∠OBC = 40°. Find ∠BAC. **2**



25. Find the second, third and fourth terms of the sequence defined by $t_1 = 7$ and $t_n = t_{n-1} - 3$ for $n \geq 2$. **2**

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Show that $0.285714285714\ldots = \frac{2}{7}$. **3**

27. 200 students were asked which type of book they borrowed: **3**

Type	Fiction	Science	History	Comics
Students	70	50	30	50

A student is chosen at random. Find the probability that the student borrowed (i) fiction, (ii) science or history, (iii) not comics.

OR

The number of goals scored by a team in 40 matches:

Goals	0	1	2	3
Matches	6	14	12	8

A match is chosen at random. Find the probability that the team scored (i) exactly 2 goals, (ii) at least 2 goals, (iii) no goal.

28. Find the coordinates of the point on the y-axis which is equidistant from A(−5, 2) and B(3, −2). **3**

29. Simplify: $(a + b)^3 - (a - b)^3$. **3**

30. A chord of a circle of radius 15 cm is 18 cm long. Find (i) its distance from the centre; (ii) the length of another chord of this circle at a distance of 9 cm from the centre. **3**

31. A triangular plot has sides 20 m, 21 m and 29 m. Find its area using Heron's formula. **3**

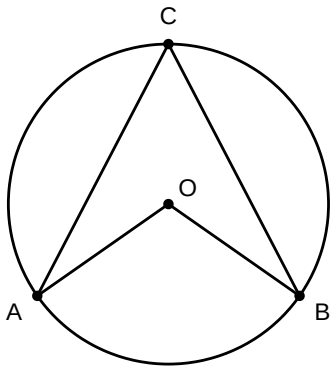
OR

Find the area of a triangle whose sides are 8 cm, 15 cm and 17 cm using Heron's formula.

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. Prove that the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the remaining part of the circle (take the case when the arc is a minor arc). **5**



33. The four corners of a flower bed are marked on a grid as A(3, 2), B(0, 5), C(−3, 2) and D(0, −1). **5**
(a) Plot the points and join them in order. (2 marks)
(b) Find the lengths AB, BC, CD and DA. (1½ marks)
(c) Find the diagonals and justify that ABCD is a square. (1½ marks)

34. How many multiples of 9 lie between 100 and 500? Find their sum. (2½ + 2½) **5**

OR

The 5th term of an AP is 17 and its 12th term is 45. Find the AP (first three terms) and its 20th term.

35. The breadth of a rectangular field is 7 m less than its length and its area is 120 m². Form a quadratic expression, factorise it to find the length and breadth, and find the perimeter. **5**

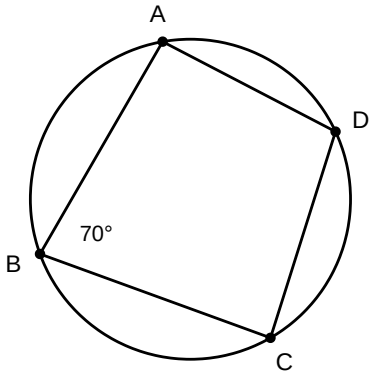
OR

The area of a rectangular carpet is $(6x^2 + 11x + 3)$ m². Factorise the expression to find its length and breadth, and find its perimeter when $x = 3$.

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A circular running track has four floodlight poles A, B, C and D on its boundary, forming a cyclic quadrilateral ABCD. ∠ABC = 70° and ∠BAD = 85°. O is the centre of the track.



(i) Find ∠ADC. 1

(ii) Find ∠BCD. 1

(iii) If arc BC subtends ∠BAC = 30° at A, find ∠BOC. 2

OR

(iii) A chord of the track is 48 m long and lies 7 m from the centre. Find the radius of the track.

37. The seats in an auditorium are arranged in rows. The first row has 12 seats and every next row has 2 more seats than the row before it. The number of seats in the n-th row is $t_n = 2n + 10$.

(i) How many seats are there in the 8th row? 1

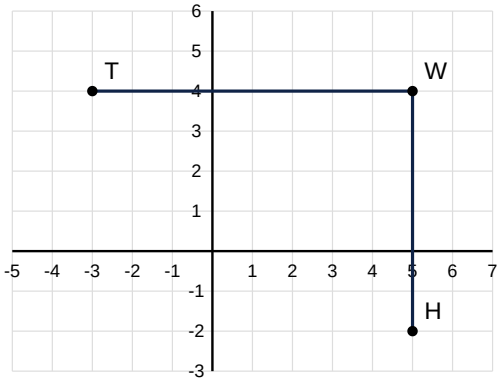
(ii) Write the common difference of the sequence of seats. 1

(iii) Which row has 40 seats? 2

OR

(iii) Find the total number of seats in the first 10 rows.

38. The map of a treasure hunt is drawn on a coordinate grid (1 unit = 10 m). The old tree is at T(−3, 4), the well is at W(5, 4) and the hut is at H(5, −2).



(i) Find the distance between the tree and the well (in units). 1

(ii) Find the midpoint of the line joining the tree and the hut. 1

(iii) Find the point on the x-axis which is equidistant from the tree and the well. 2

OR

(iii) Find the straight-line distance between the tree and the hut.